

Intestinal iron absorption: regulation by dietary & systemic factors.

.

Iron is an essential trace metal in human metabolism. However, imbalances in iron homeostasis are prevalent worldwide and have detrimental effects on human health. Humans do not have the ability to remove excess iron and therefore iron homeostasis is maintained by regulating the amount of iron entering the body from the diet. Iron is present in the human diet in number of different forms, including heme (from meat) and a variety of non-heme iron compounds. While heme is absorbed intact, the bioavailability of non-heme iron varies greatly depending on dietary composition. A number of dietary components are capable of interacting with iron to regulate its solubility and oxidation state. Interestingly, there is an emerging body of evidence suggesting that some nutrients also have direct effects on the expression and function of enterocyte iron transporters. In addition to dietary factors, body iron status is a major determinant of iron absorption. The roles of these important dietary and systemic factors in regulating iron absorption will be discussed in this review.